

Akula Vinay

Email: vinayusing@gmail.com

Phone: +91-8885666263

Objective

To be an associate with a progressive organization that gives me scope to update my knowledge and skills in association with latest technologies and a part of a team that dynamically works towards growth of organization and gain satisfaction.

Educational Qualification

- Graduated from Joginpally BR Engineering College , Hyderabad in 2015
- Intermediate from SR Junior College , Warangal
- SSC from Tawakkal High School , Mancherla

Professional Summary

- Carrying 8+ years of overall experience in the field of AI.
- Used CNN toolkit for easy porting with Caffe, TensorFlow, and ONNX
- Proficiency in Statistical Model, Machine learning algorithms to support business evolution.
- Abilities in handling multiple priorities, with a bias for action and a genuine interest in personal and professional development.
- Strong exposure on **Python, Django, OpenCV, Neural networks, GENAI, Agentic AI, VLLM, LLM.**
- Good communication & interpersonal skills with proven abilities in resolving complex software issues
- Ability to work effectively and interact with other members in the team
- Very well organized with demonstrated ability to work under pressure, prioritize work and meet tight deadlines
- Developed an end-to-end AI based application based on client requirements.
- Deployment of Tensorflow Models using TF serving.
- Building **C/C++ (STL , Factory Design Pattern)** Applications for Arm processors.
- Cross Compilation for ARM using Cmake and Makefile.
- Developed a Production level CNN object detection in TensorFlow Lite for Microcontrollers.
- Familiarity in AWS IOT Core with IoT communication protocols like MQTT
- Image processing algorithms and Edge device porting experience.

Technical Expertise

Python, Django, MongoDB, MySQL, Machine Learning, Deep learning, OpenCV, Neural networks, Tensorflow, Pytorch, keras, Object detection, Image Classification, Image Processing.

Edge Platforms

Ambarella cv2, Ambarella cv52 co, Ambarella cv28m cashew nut, Ambarella cv22 Walnut , Ambarella cv25 Hazelnut , Nvidia Jetson nano , Qualcomm RB5 , Intel NUC , NXP

Professional Experience

- Working As Senior Specialist Data sciences -IIOT in **LTIMINDTREE**, From 07 Feb to Present
- Worked as Machine Learning Engineer in **DeepEdge AI India Private Limited**, From 03 August 2020 to 23 December 2022
- Worked as **Data Science** associate in **Digitele Networks Private Limited**, from 01 January 2019 to July 2020.
- Worked as a **Data analyst** at **Sears Holdings Corporation**, Hyderabad from 04 April 2017 to 13 December 2018.
- Worked as an **Intern** at **ValueLabs**, Hyderabad from 18 April 2016 to 25 August 2016

Project-07 : Bearing Wear Inspection

Client: Honda

Analytical Tools & Environment: Python,VLLM,, LLM, Computer Vision.

Duration: June 2024 till date

Description :

Developed a LLM based model to summarize and create a maintenance report along with Computer vision model to identify type of defects and deploy it in production following Mlops cycle using AWS services.Object detection model is built to identify wear type.Metadata information extracted from above model is used for template creation. LLM model like GPT3.5 is used to analyze the data to create structured data which is then compared with the above template to provide maintenance and summary report.

Project-06: SDK Development for Multiple Edge devices.

Client : Product

Job Role : Developed an Evaluation Application

Softwares On GPU : Python, C/C++ Programming , Pytorch , Mxnet , Tensorflow, ONNX ,CV and other Machine Learning related Libraries, AWS-IOT-CORE, S3, Cross compilation

Description:

Prepare the Conversion Scripts for Multiple Models for multiple edge devices and create payload. Evaluation application using Factory Design Pattern runs inference to multiple models in multiple edge devices. Connect Multiple edge devices to AWS IOT CORE with different topics and communicate with the device to run the evaluation and send the results to S3 Storage.

Project-05: Model Compression and Quantization For Porting On Edge.

Client : Product

Job Role : Developed an Neural Network Models for Detection and Applying Compression and Quantization Techniques

Softwares On GPU : Python, C/C++ Programming , Pytorch , Mxnet , Tensorflow, ONNX ,CV and other Machine Learning related Libraries.

Description:

Trained Tensorflow and Pytorch models for Publicly Available Datasets and Applied Compression and Quantization to improve the Performance and Accuracy in Qualcomm rb5 , Xilinx ,Ambarella cv25 EVK Development boards.

Project-04: Face Detection and Face Recognition On Edge.

Client : Confidential

Job Role : Developed Neural Network Models for Face Detection and Face Recognition and Deployed in Battery Based Edge computing Device For Smart Policing .

Softwares On GPU : Python, C/C++ Programming , Pytorch , Mxnet , Tensorflow, ONNX ,CV and other Machine Learning related Libraries.

Description:

Explored and tested several face detection and face recognition architectures on GPU to zero in on models having best accuracy and speed. Created a pipeline for video face recognition consisting of face detection (FD), face alignment (FA), face tracking (FT) and face recognition (FR)

Project-03:Collecting benchmarking from Ambarella Edge Device.

Client : Safely-you

Job Role : Computer Vision Engineer

Framework : PyTorch/Onnx

Description:

Customers based on their requirements identified a few pretrained models and wanted to evaluate the models on Edge Device Ambarella CV22/CV25 to collect the benchmarking parameters like performance..

Responsibilities:

- Model conversion from PyTorch to ONNX and further device compatibility model generation.
- Developed complete inference application running on CV22/CV25 with live camera and static images.
- Inference pipeline setup, pre-processing, post-processing on ARM.

Project-02: Detecting defective solar panels and wind blades using Drone images.

Client : Greenko

Job Role : Developed an Django application and build object detection and image classification models using Deep learning and deploy in the production.

Technology : Python, Keras, Tensorflow, Deep learning, Machine learning, image processing algorithms, MySQL.

Description:

Solution avoids the manual inspection of solar panels and wind blades by using IR camera enabled drones and RGB camera enabled drones respectively and detect anomalies (dirt, shade, glass breakage, etc., in solar panels and , laminate damage, detachment of collar blade, paint erosion etc., in WTG blades).

Project-01: Automatic Number Plate Recognition (ANPR)

Client : Ace Urban

Job Role : Developed an Django application and build object detection models using Deep learning and deploy in the production.

Technology : Python, Tensorflow, OpenCV, deep learning, Machine learning ,image processing algorithms, MySQL

Description:

Automatic Number Plate Recognition (ANPR) is a technology that uses advanced image processing and optical character recognition on RTSP (IP) camera live streams/images to read vehicle registration plates in real time to create vehicle location data.